

Howard Tuan

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GitHub: github.com/howardtuan

Skills

- Languages: Python, Java, C
- Databases: MySQL, PostgreSQL
- Machine learning and Deep learning: Tensorflow, PyTorch, Scikit-Learn
- Tools: Docker, Git

Education

National Chung Hsing University, MS in Information Management Sept 2024 –Jul 2026

- Graduation Ranking: 4/45 | Average Grade: 92.4 | GPA: 4.21/4.3
- Thesis: Deep Learning-Based Detection, Segmentation, and Classification of Thyroid Nodules in Ultrasound Images

Chung Yuan Christian University, BS in Information Management Sept 2020 –Jul 2024

- Graduation Ranking: 12/113 | Average Grade: 88.31 | GPA: 3.92/4
- Graduation Project: Talk to Draw

Experience

Programming Instructor, Orange Apple Programming School Dec 2021 –Present

- Taught APCS certification courses and guided students in Scratch, Python, JavaScript, HTML, Database, Algorithm, and AI to strengthen programming foundations and problem-solving skills

Teaching Assistant, Precision Medicine and Clinical Translation Research Lab, NCHU Sep 2024 –Jul 2026

- Assisted with course design, assignment grading, and Python tutoring for Data Mining and Big Data Analysis and Applications

Lecturer, AI and Data Science Workshop, NCHU Affiliated Senior High School Dec 2025 & Jan 2026

- Designed hands-on Python lessons and guided high school students in learning data mining and basic machine learning concepts

AI Intern, CanCell Technology Co., Ltd. Feb 2026 –Mar 2026

- Assisted in developing a wound image recognition model and integrated deployed AI model services from the Insight platform into a LINE Bot

Software Engineering Intern, Cathay Life Insurance Feb 2024 –Jun 2024

- Developed a reverse engineering tool to convert Java methods into SPEC documents and gained experience in software development and system maintenance

Projects

Local AI Teaching Assistant System - MOE Teaching Practice Research Program Aug 2025

- Built a local RAG-based Q&A system with OpenWebUI for NCHU's Big Data Analysis and Applications course
- Enabled students to query course materials, assignment requirements, and learning questions, improving after-class learning support

LightRAG Harry Potter Novel Query System Mar 2025

- Built a LightRAG-based query system using a local language model and embedding model on the original English Harry Potter novels
- Applied graph-based retrieval to improve reasoning over complex character relationships and storylines

Stock Prediction and Factor Investment Analysis Using CNN Nov 2024 –Mar 2025

- Collaborated with NCHU Finance Department on an interdisciplinary research project
- Built and evaluated CNN models for stock trend and factor-based indicator analysis; results were published as an academic paper

Certification

TOEIC 745

